

Scope of You: Supplementary Appendix

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A Diagnostic objective settings

The controlled PCO optimizer uses reward, confidence-scaled trust, supervised anchoring, user-state intervention, context selection, protected preservation, and invariant-attack terms. The main robust runs use the same fixed objective weights across five seeds. Diagnostic ablations and the loss-scale frontier use a smaller fixed training budget and are not mixed with the headline five-seed estimate.

B Proof of the average leakage bound

For each protected behavior dimension d , let $\Delta_d = b_d(x, I_u(u)) - b_d(x, u)$. If $\mathbb{E}[\Delta_d^2] \leq \epsilon_d^2$, then by Cauchy-Schwarz,

$$\mathbb{E}|\Delta_d| \leq \sqrt{\mathbb{E}[\Delta_d^2]} \leq \epsilon_d. \quad (1)$$

Linearity of expectation then gives

$$\mathbb{E} \left[\frac{1}{|P|} \sum_{d \in P} |\Delta_d| \right] \leq \frac{1}{|P|} \sum_{d \in P} \epsilon_d. \quad (2)$$

The same argument yields an average absolute bound for the squared suppress loss around a neutral target. These propositions certify only the explicitly measured coordinates.

C Controlled compute disclosure

A representative 180-step, batch-512 headline optimization run using the frozen tensor cache required 7.49 seconds wall time on a five-vCPU AMD EPYC 7763 runtime, with 345,880 KiB peak resident memory and no accelerator. Five headline optimization seeds therefore require roughly 37.5 seconds of model-optimization wall time with cached tensors; preprocessing, evaluation, ablations, failed experiments, and artifact generation are additional. No GPU-hour estimate is reported because the planned 7-8B study has not been executed.